

Crowd Density Estimation using VGG16-based CSRNet

Summary of IDA Journal Article (2025)

1. Problem Statement

- Crowd density estimation aims to determine the number of people and their spatial distribution in an image.
- It is challenging due to occlusion, scale variation, background clutter, and dense scenes.
- Accurate estimation is important for surveillance, public safety, and urban planning.

2. Proposed Approach

- The paper proposes a deep learning-based solution using a Convolutional Neural Network (CNN).
- VGG16 is used as a **front-end** feature extractor.
- **CSRNet** is used as a **back-end** network with dilated convolutions.
- The model generates a density map instead of directly predicting a single crowd count.

3. Working of the Model

- Input crowd images are first passed through VGG16 convolutional layers.
- These layers extract low-level and high-level visual features.
- The extracted features are processed by CSRNet.
- The final output is a crowd density map, whose integral gives the total crowd count.

4. Datasets and Evaluation

- The model is evaluated on the Mall dataset and the Crowd-UIT dataset.
- Performance is measured using MAE and Mean Squared Error.
- Lower MAE and MSE values indicate higher prediction accuracy.

5. Results and Applications

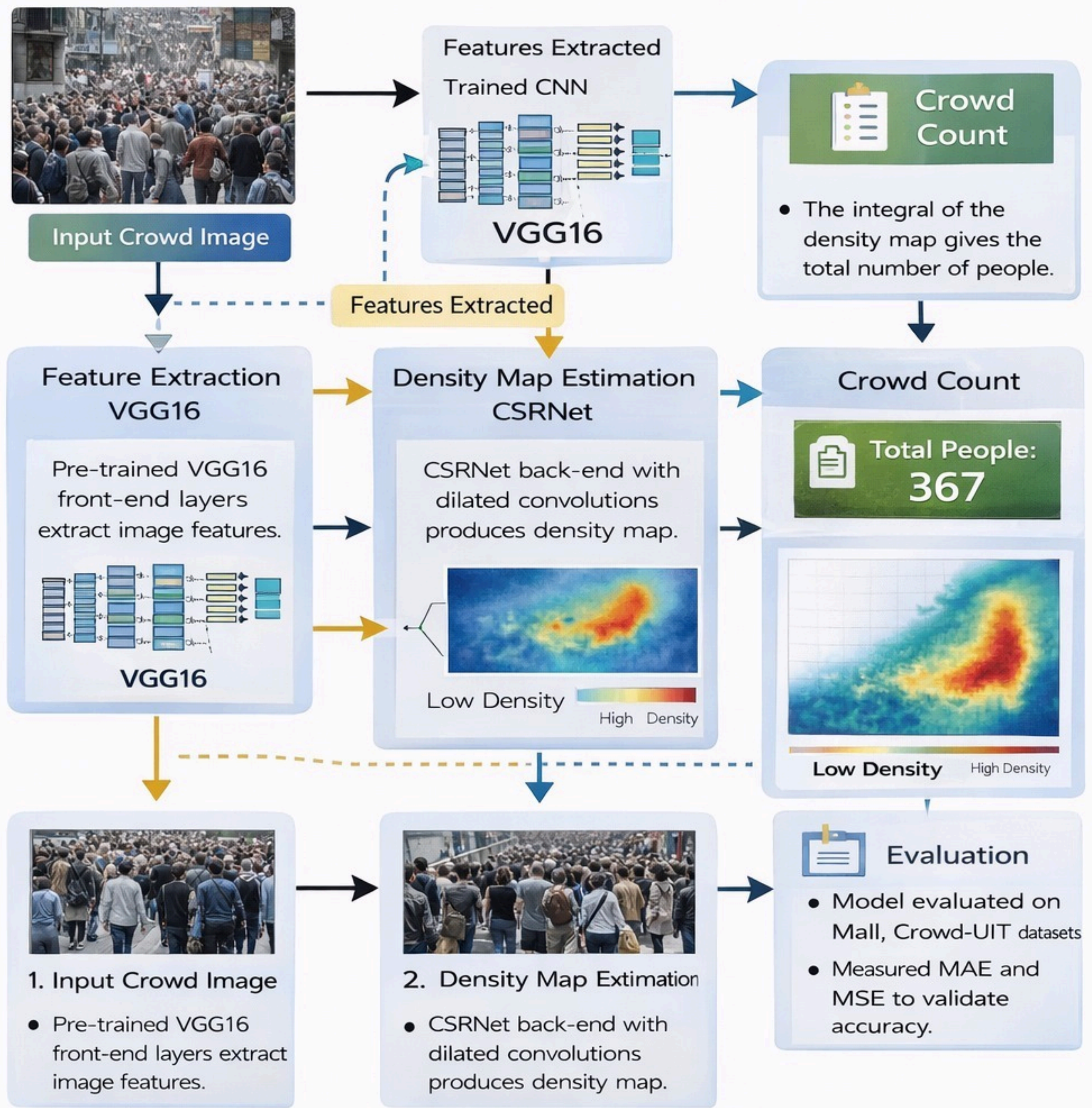
- The proposed model achieves low error values on both datasets.
- It performs well across different indoor and outdoor environments.
- Applications include:
 - Crowd monitoring
 - Public safety management
 - Event planning
 - Smart surveillance systems

6. Conclusion

- The VGG16-based CSRNet model effectively estimates crowd density.
- Density maps provide more informative results than simple crowd counts.
- The approach is suitable for real-world crowd analysis applications.

Crowd Density Estimation using VGG16-based CSRNet

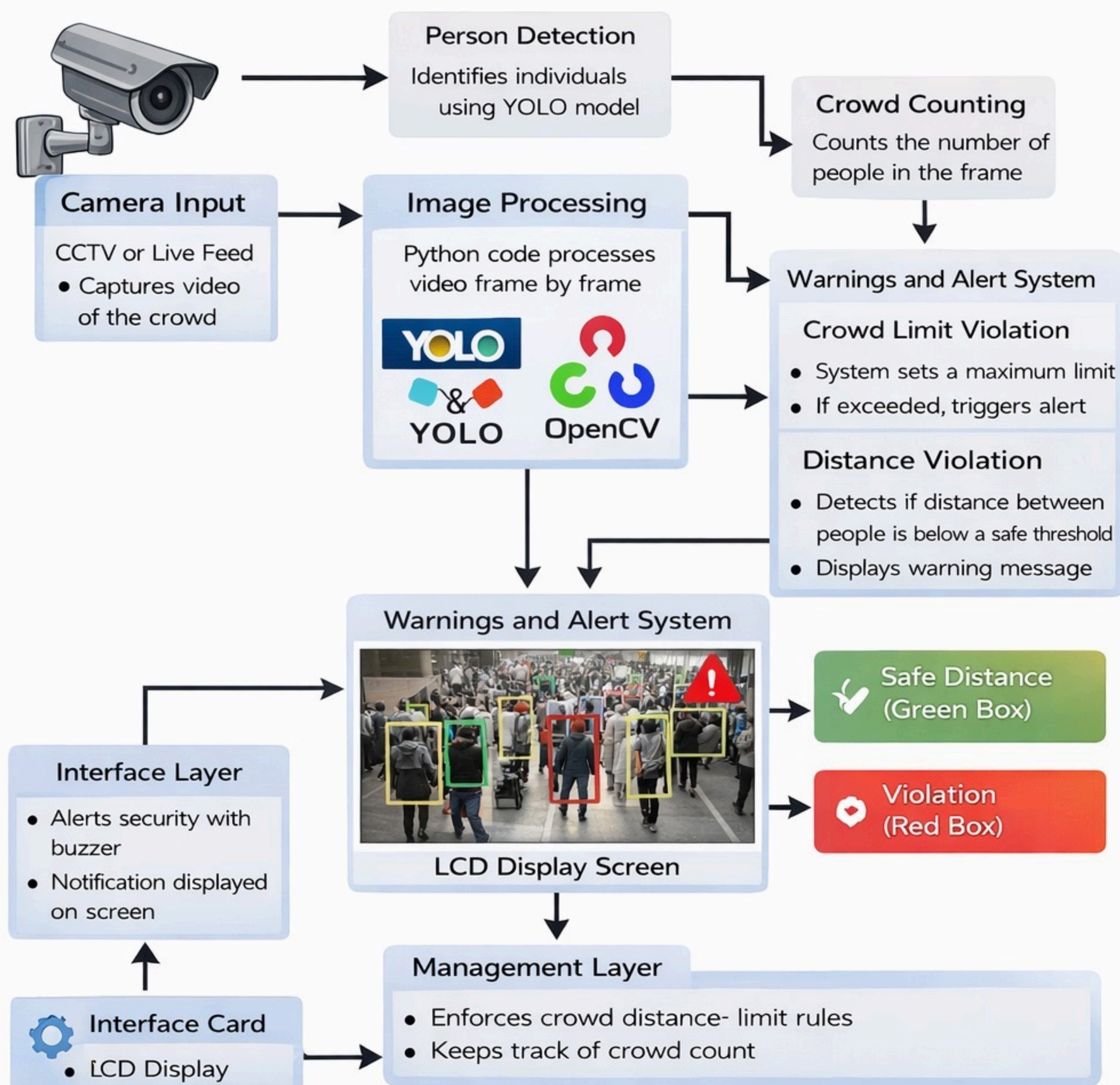
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Key Points

- Combines pre-trained VGG16 for feature extraction and CSRNet for density estimation.
- Generates a density map highlighting high-density areas.
- Provides accurate crowd counts by integrating the density map.
- Evaluated with low error rates on benchmark Mall and Crowd-UIT datasets.

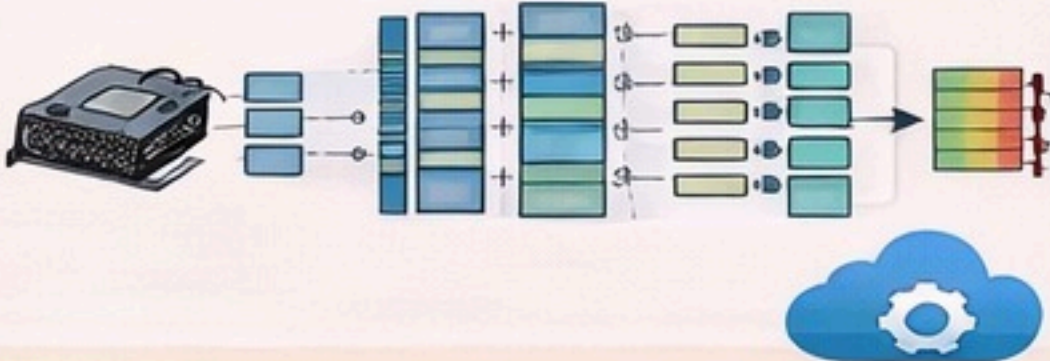
Crowd Monitoring System Using Image Processing



Working of the Model

- CCTV captures video of the crowd, YOLO & OpenCV detect people, and a warning alert is triggered if crowd limits or safe distances are violated.
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Cost Comparison Between Crowd Estimation Models

YOLO-based Monitoring		VGG16 + CSRNet	
			
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	Hardware Cost	- Low-Medium	Density-based
	GPU Requirement	✓	Mandatory
	Cloud Dependency	No	Often Yes 
	Software Cost	Free	Free
	Training Cost		High
	Maintenance		Complex
₹ 45,000 – ₹ 90,000		₹ 1,20,000 – ₹ 2,00,000+	
✓ Affordable		✗ Expensive	

Low Cost	High Cost
₹ 45,000 – ₹ 90,000	₹ 1,20,000 – ₹ 2,00,000+
Affordable	Expensive
₹ Low Cost	₹ High Cost

Comparison of Crowd Estimation Methods

YOLO-based Crowd Monitoring



ADVANTAGES

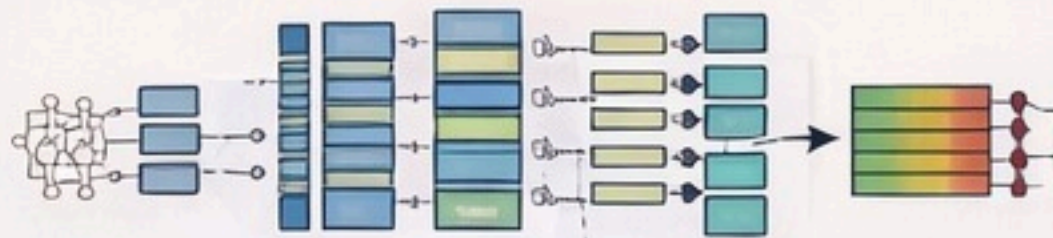
- Real-time person detection
- Distance & limit monitoring
- Good for sparse/medium crowds

ADVANTAGES

- Not suitable for dense crowds
- Can miss people in large gatherings, like festivals,...

VS

VGG16-based CSRNet



DISADVANTAGES

- More accurate count in dense crowds
- Produces informative density maps

DISADVANTAGES

- Slower, not suitable for live monitoring
- No individual person details

Features	YOLO-based Crowd Monitoring	VGG16-based CSRNet
Approach	Detection-based	Density-based
Good For	Sparse/medium crowds	Dense/large crowds
Detects Individuals	✓ Yes	✗ No
Real-time Monitoring	✓ Yes	✗ No